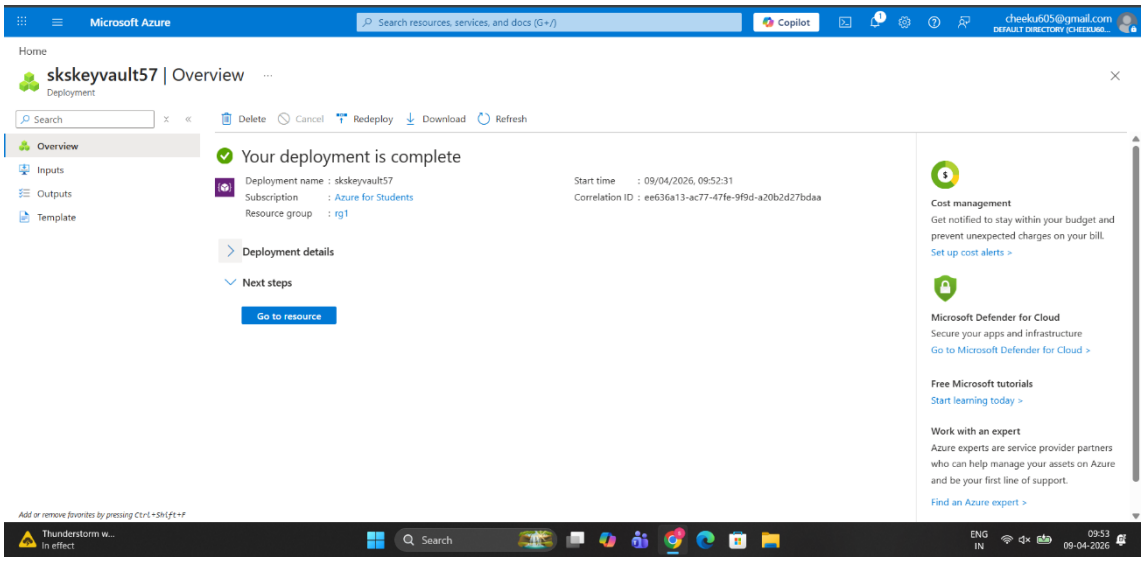
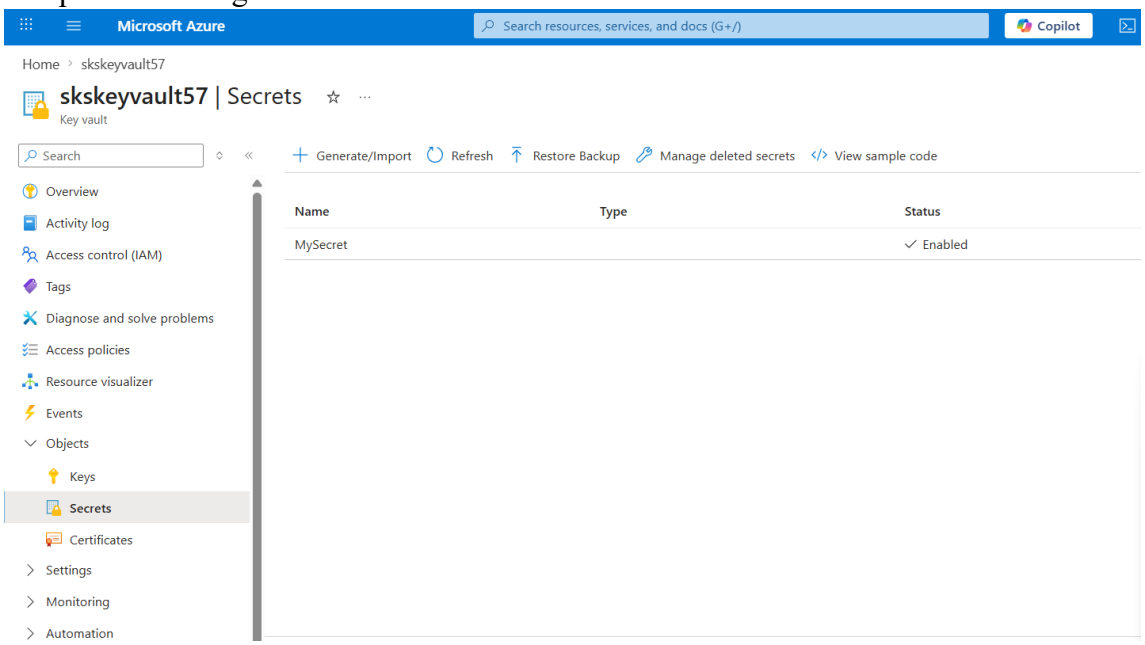
Day - 78

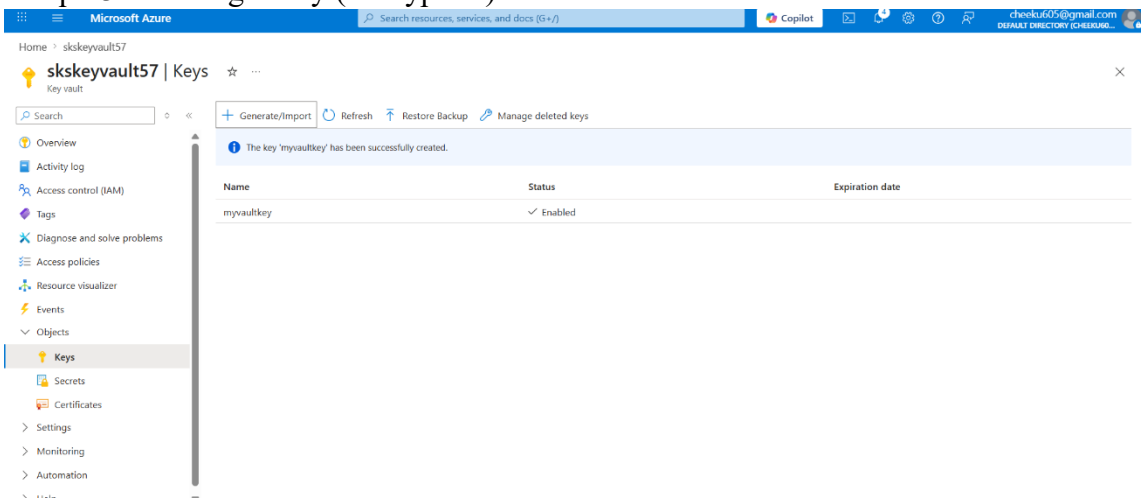
Step - 1: Create the Key Vault



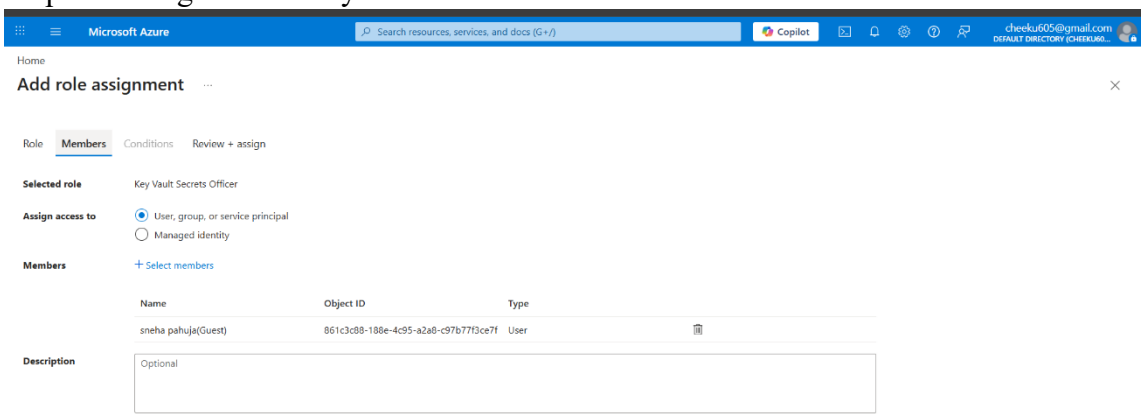
Step - 2: Granting Data Access

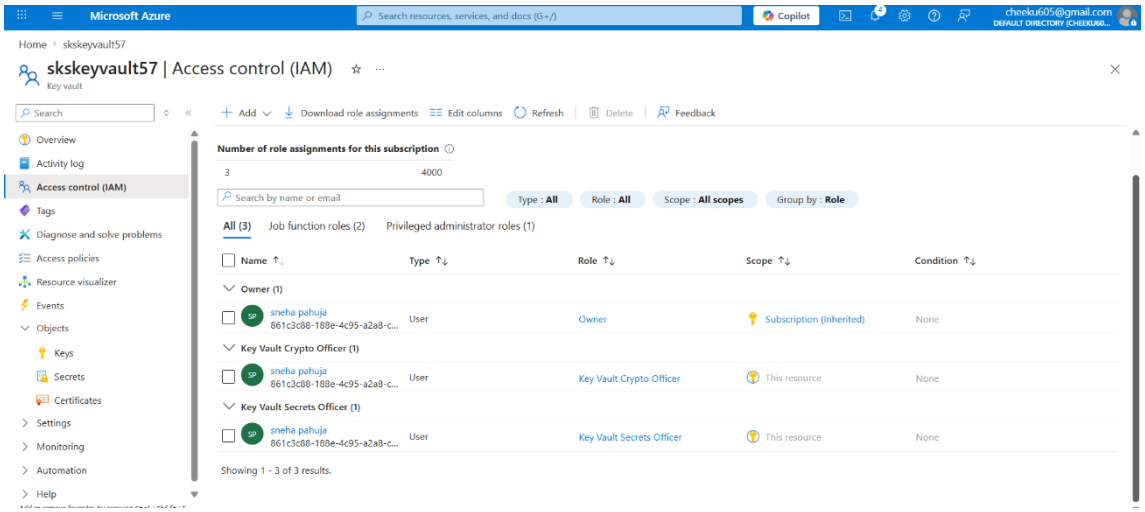


Step - 3: Creating a Key (Encryption)



Step - 4: Assign Roles To your User





Step - 5: The Implementation Code

```
ConsoleAzureDemo | ConsoleAzureDemo.Program | Main(string[] args)

using Azure.Identity;
using Azure.Security.KeyVault.Keys.Cryptography;
using System.Text;

namespace ConsoleAzureDemo
{
    0 references
    internal class Program
    {
        0 references
        static void Main(string[] args)
        {
            string keyVaultUrl = "https://crazzyy-vault.azure.net/";
            string keyName = "DataEncryptionKey";

            var credential = new DefaultAzureCredential();

            var cryptoClient = new CryptographyClient(
                new Uri($"{keyVaultUrl}keys/{keyName}"),
                credential
            );

            string plainText = "Hello SmartBank!";
            byte[] data = Encoding.UTF8.GetBytes(plainText);

            // Encrypt
            var encryptResult = cryptoClient.Encrypt(
                EncryptionAlgorithm.RsaOaep,
                data
            );

            Console.WriteLine($"Encrypted: {Convert.ToBase64String(encryptResult.Ciphertext)}");

            // Decrypt
            var decryptResult = cryptoClient.Decrypt(
                EncryptionAlgorithm.RsaOaep,
                encryptResult.Ciphertext
            );

            string decryptedText = Encoding.UTF8.GetString(decryptResult.Plaintext);
            Console.WriteLine($"Decrypted: {decryptedText}");

90 %   No issues found   Ln: 12, Ch: 48   SPC   CRLF   UTF-8 with BOM
```

Output

```
Microsoft Visual Studio Debu

Encrypted: m+bU/d9PbsLlTh2hQiT5jvXH5JNpwvtiGOuS18D4lgtcxRMQRwyn+J+Z2DO4Fzj7gNpmK0TcpvlqDz0/ixxRHKI32HiUgUyOFmiN/ZIUsG8F+
QFQureK5QpTPxh/41dc4kaGBQsbo6DchDUDQmGpS9TNjsdgNTbTJB/z3M2QqHMJC8hXfDuw8daL0RmiG5csOPvnK/tT7wFHAjs5DUuf36Vyd3gjsN2uFSA6j
+uTyAV1Wle+0HP8oacovnTWTav8v1xQnnHRSa6JhEshmJWFoGYGKgzt/wMm9t3hHDer1yYbPH7OXZzVd1abAV/VqAks1XehutP6XpMLSWWDnObu4w==
Decrypted: Hello SmartBank!
```